

TASK 3 REPORT

Title

Exploratory Data Analysis (EDA) on Netflix Dataset

Objective

The objective of this task is to perform Exploratory Data Analysis (EDA) on the Netflix dataset to identify trends, patterns, and insights related to movies and TV shows available on the Netflix platform.

Tools and Technologies Used

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Jupyter Notebook

Dataset Description

The Netflix dataset contains information about movies and TV shows available on Netflix, including title, type, country, release year, rating, duration, director, cast, and genre.

Methodology

Step 1: Data Loading

The dataset was imported using the Pandas library and examined to understand its structure.

Step 2: Data Inspection

The following operations were performed:

- Displayed first few rows using head()
- Checked dataset information using info()
- Generated statistical summary using describe()

Step 3: Data Cleaning

- Handled missing values
- Removed unnecessary columns if required
- Checked for duplicate records

- Standardized data formats

Step 4: Exploratory Data Analysis

Several visualizations were created to analyze the dataset:

Movies vs TV Shows Analysis

Compared the number of Movies and TV Shows available on Netflix.

Country-wise Content Distribution

Identified countries contributing the most content.

Rating Analysis

Analyzed the distribution of content ratings such as TV-MA, TV-14, PG, and others.

Release Year Analysis

Examined how content production changed over time.

Genre Analysis

Investigated the most popular genres available on Netflix.

Step 5: Visualization

Graphs and charts were created using Matplotlib and Seaborn to represent insights clearly.

Results

The analysis revealed several interesting patterns:

- Movies constitute a larger portion of Netflix content compared to TV Shows.
- The United States contributes the highest number of titles.
- Most content is targeted toward mature audiences.
- Significant growth in content production has occurred in recent years.
- Certain genres dominate the Netflix catalog.

Observations

- Netflix has expanded its content library rapidly.
- Content diversity has increased over time.
- Viewer preferences influence content categories and ratings.

Conclusion

Exploratory Data Analysis helped uncover valuable insights about Netflix content distribution, ratings, genres, and release trends. The findings can support data-driven decision-making and content strategy planning.

Learning Outcomes

- Understanding Exploratory Data Analysis techniques.
- Data cleaning and preprocessing.
- Creating meaningful visualizations.
- Extracting insights from real-world datasets.
- Using Python libraries for data analysis and visualization.